

# Question Paper

Exam Date & Time: 30-Apr-2025 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech IV Semester

End Semester Examination - April / May -2025

Digital Signal Processing [ECM-2221]

Marks: 50

Duration: 180 mins.

### PART- A

Answer all the questions.

- 1) Determine the Z-transform of  $u[n - 2]$  and its Region of Convergence? (2)

A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) State and prove the initial value theorem of one sided z-transform. (3)

Course Outcome: "CO1" / Cognitive Level: "Remember"

- C) Determine the non-causal signal  $x(n]$  having the z-transform (5)

$$x(z) = \frac{1}{(1 - 2z^{-1})(1 - z^{-1})^2}$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) State the Nyquist sampling theorem. Why is it important? (2)

- A) Course Outcome: "CO2" / Cognitive Level: "Remember"

- B) Compute the 6-point DFT of the sequence  $x(n) = \{0, 1, 2, 3, 2, 1\}$ . (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Determine the response of the discrete time LTI system described by the following difference equation for the unit step input. The system has an initial condition  $y(-1) = \frac{2}{9}$ . (5)

$$y(n) = -0.8 y(n - 1) + x(n)$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3) (5)

A)

Use unilateral z-transform to find the response for the system described by the difference equation  $y(n) = 0.5 y(n - 1) + x(n - 1)$  with  $x(n) = 0.5^n u(n)$  and  $y(-1) = 1$ .

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Compute 8-point DFT of the sequence  $x(n) = \{3, 2, 1, 0, 1, 2\}$  using decimation-in-frequency FFT algorithm. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4) List the differences between FIR and IIR filters. (2)

- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Realize the following IIR filter using direct form-I and direct form-II. (3)  
 $y(n) = -0.1y(n - 1) + 0.2y(n - 2) + 3x(n) + 3.6x(n - 1) + 0.6x(n - 2)$

Course Outcome: "CO3" / Cognitive Level: "Understand"

- C) Determine the impulse response of FIR filter of length 7 to meet the following specifications. (5)

$$H_d(e^{j\omega}) = \begin{cases} e^{-j(M-1)\frac{\omega}{2}} & \text{for } 0 \leq \omega \leq \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

Use rectangular window.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Bring out the salient features of parametric PSD estimation. (2)

- A) Course Outcome: "CO5" / Cognitive Level: "Remember"

- B) Use impulse invariant transformation to convert the analog filter with system function (3)

$$H(s) = \frac{1}{(s + 0.5)(s^2 + 0.5s + 2)}$$

into a digital filter. Select  $T=1$  s.

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Compute the minimum order and transfer function  $H(s)$  of a Chebyshev type-I low pass filter, where the maximum allowable ripple is 1 dB in the pass band extending from 0 to  $0.1\pi$  radians/sec. The minimum attenuation should be 40 dB at the stop band edge frequency of  $0.3\pi$  radians/sec. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

-----End-----

| Event Name                                                                                   |                |             | Subject                            |
|----------------------------------------------------------------------------------------------|----------------|-------------|------------------------------------|
| MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025 |                |             | ECM-2221:Digital Signal Processing |
| #                                                                                            | Course Outcome | Description | Marks                              |
| 1                                                                                            | 1              | CO 1        | 18                                 |
| 2                                                                                            | 2              | CO 2        | 12                                 |
| 3                                                                                            | 3              | CO 3        | 5                                  |
| 4                                                                                            | 4              | CO 4        | 13                                 |
| 5                                                                                            | 5              | CO 5        | 2                                  |
| Total                                                                                        |                |             | 50                                 |

\* **NA** - CO Not Selected

| Event Name                                                                                   |             |                 | Subject                            |
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| #                                                                                            | Level       | No of Questions | Marks (%)                          |
| 1                                                                                            | Remember    | 3               | 7(14%)                             |
| 2                                                                                            | Understand  | 1               | 3(6%)                              |
| 3                                                                                            | Application | 9               | 38(76%)                            |
| 4                                                                                            | Analysis    | 1               | 2(4%)                              |
| 5                                                                                            | Create      | 0               | 0(0%)                              |
| 6                                                                                            | Evaluate    | 0               | 0(0%)                              |
| Total                                                                                        |             | 14              | 50 (100%)                          |